

Control Barrier Functions in Semi-Automated Beverage Blending Equipment

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INTRODUCTION & MOTIVATION



Fig 1: Multiplex BIC Machine¹

Quick-Service Restaurants (QSRs) are known for providing fast service but keeping up with growing customer demand to **be faster** while meeting internal goals to **decrease labor costs** while constantly expanding their offerings can only be accomplished by one thing, **automation**.

An easy way to introduce **automation** is to move away from traditional blenders to **Blend-In-Cup (BIC)** machines to create smoothies, frappes, and another blended beverages.

BIC machines need to operate **safely** around employees and need to work with a **variety of cup sizes**, for both domestic and international market applications.

By implementing a **CLF-CBF Filter**, we have the ability to apply **safety-critical controls** to food-service equipment.

PROBLEM STATEMENT

Objective:

Develop a controller that **safely guides** an immersion blender through a **predefined blending profile** while adapting to **various cup heights**.

Safety Requirements:

- Have the immersion blender **slowly approach** and dip into the cup
- Prevent** the tip of the blender from **hitting the bottom** of the cup or **exiting** the cup during blending
- Have the blender **slowly retract** out of the cup, returning to its home position at the top of the blend chamber

METHODS

PD Controller (Initial CLF)

- Drive blender tip toward desired blending trajectory
- Ensure convergence to desired trajectory and stability

CLF-CBF Filter

- Enforce barrier constraints: top & bottom
- Enforce actuator limits to dampen acceleration spikes

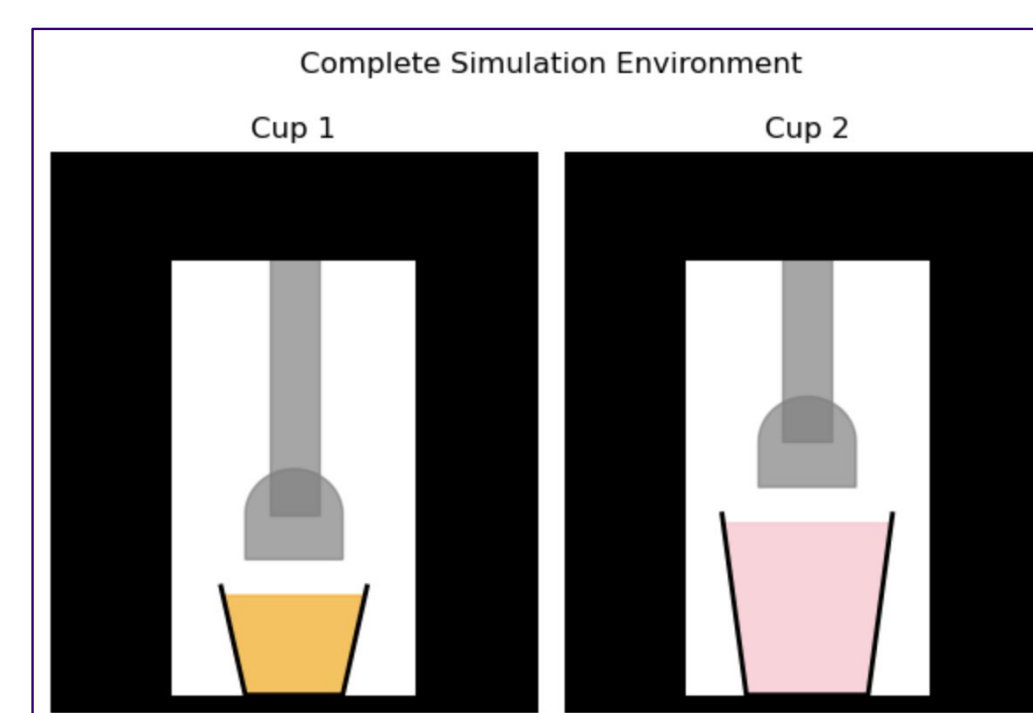


Fig 2: Complete Simulation Environment with Various Cup Sizes

SYSTEM MODEL

$$\text{dynamics} \quad \dot{x} = \begin{bmatrix} \dot{y} \\ \dot{y} \end{bmatrix}$$

$$\text{controls} \quad u = [\ddot{y}]$$

$$\text{top constraint} \quad h_t = y_{top} - y_{tip}$$

$$\text{bottom constraint} \quad h_b = y_{tip} - y_{bottom}$$

$$\text{control bounds} \quad u_{lower} \leq u \leq u_{upper}$$

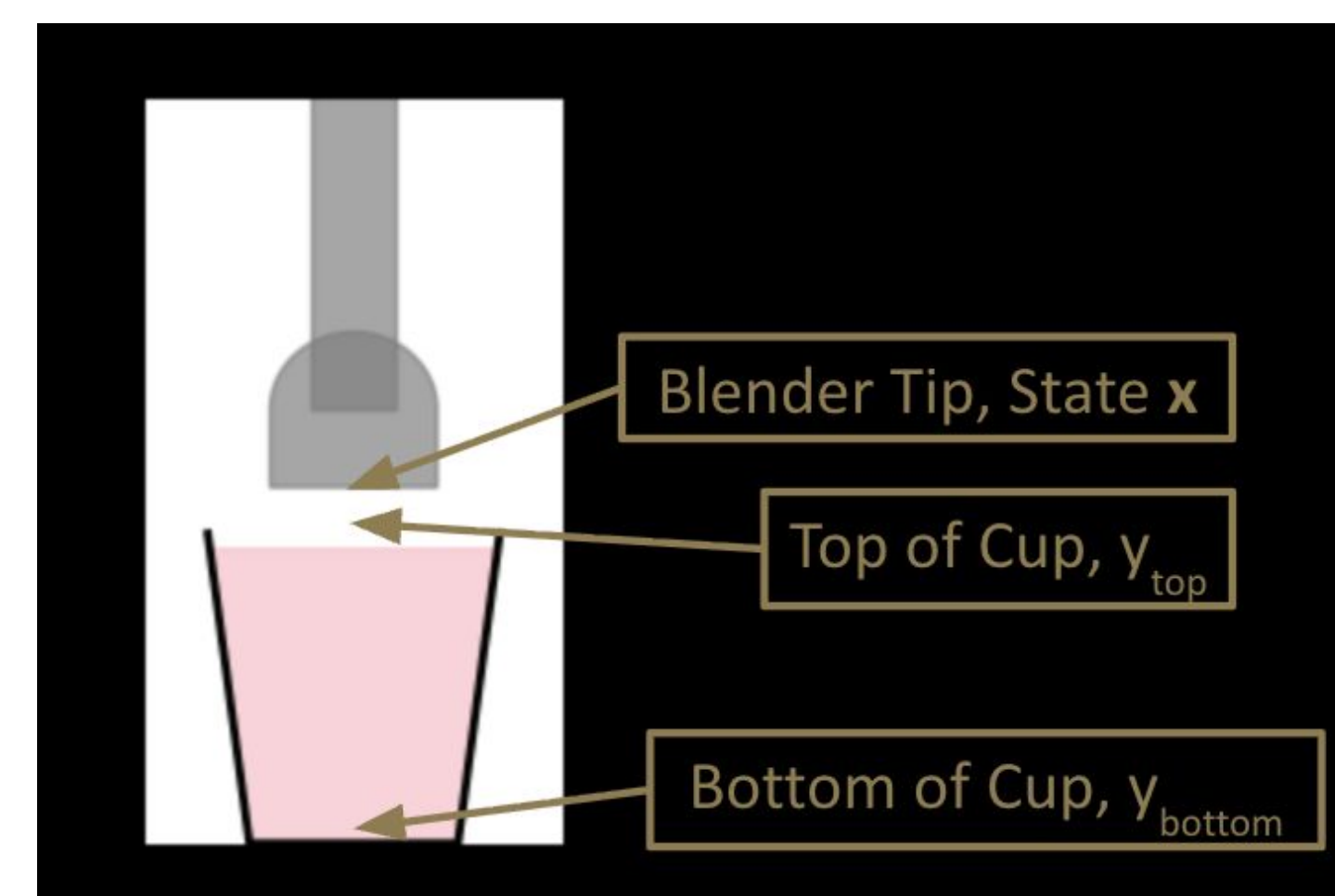


Fig 3: Diagram of System

SIMULATION 1: PD CONTROLLER

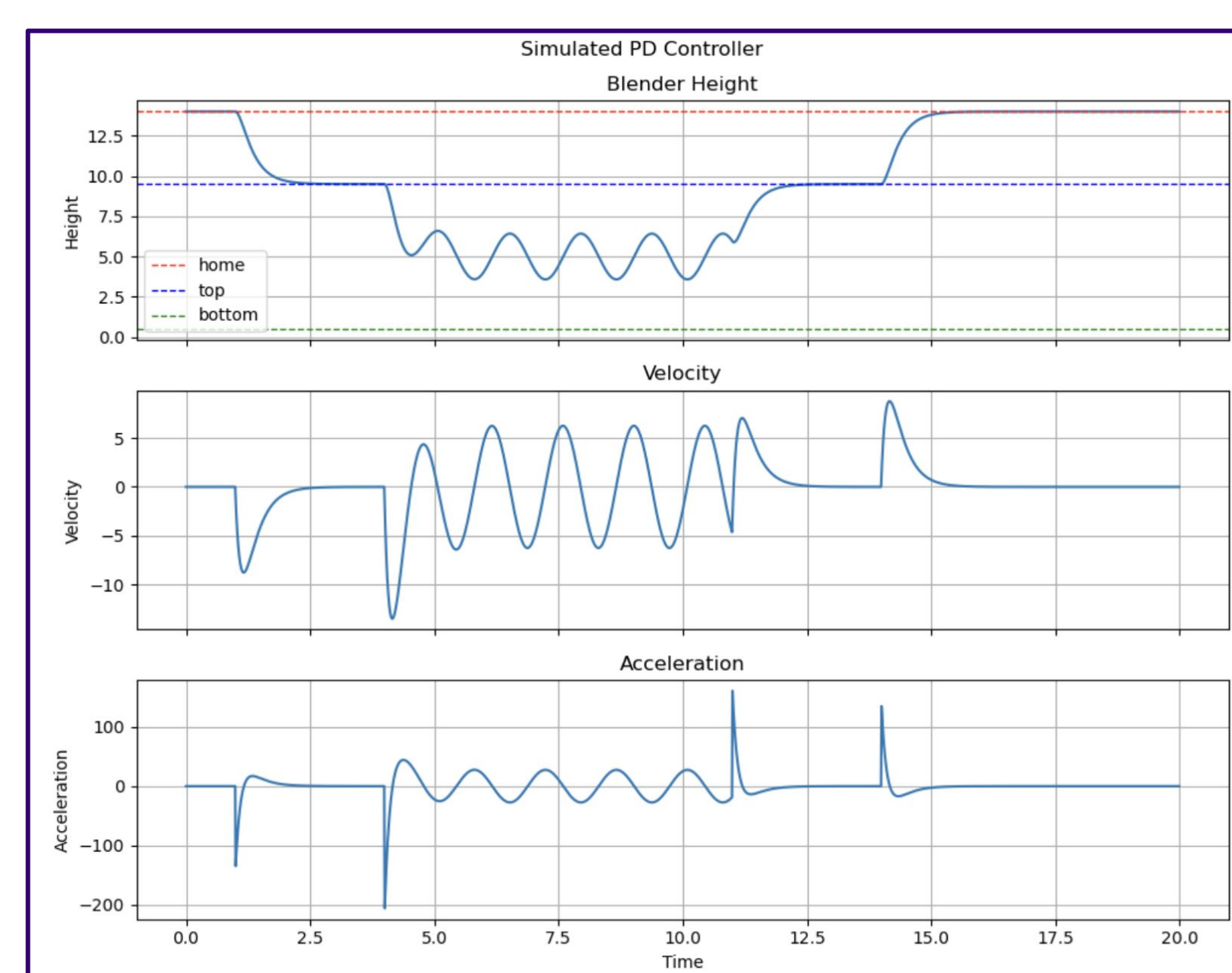


Fig 4: PD Controller Simulated Results

SIMULATION 2: CLF-CBF FILTER

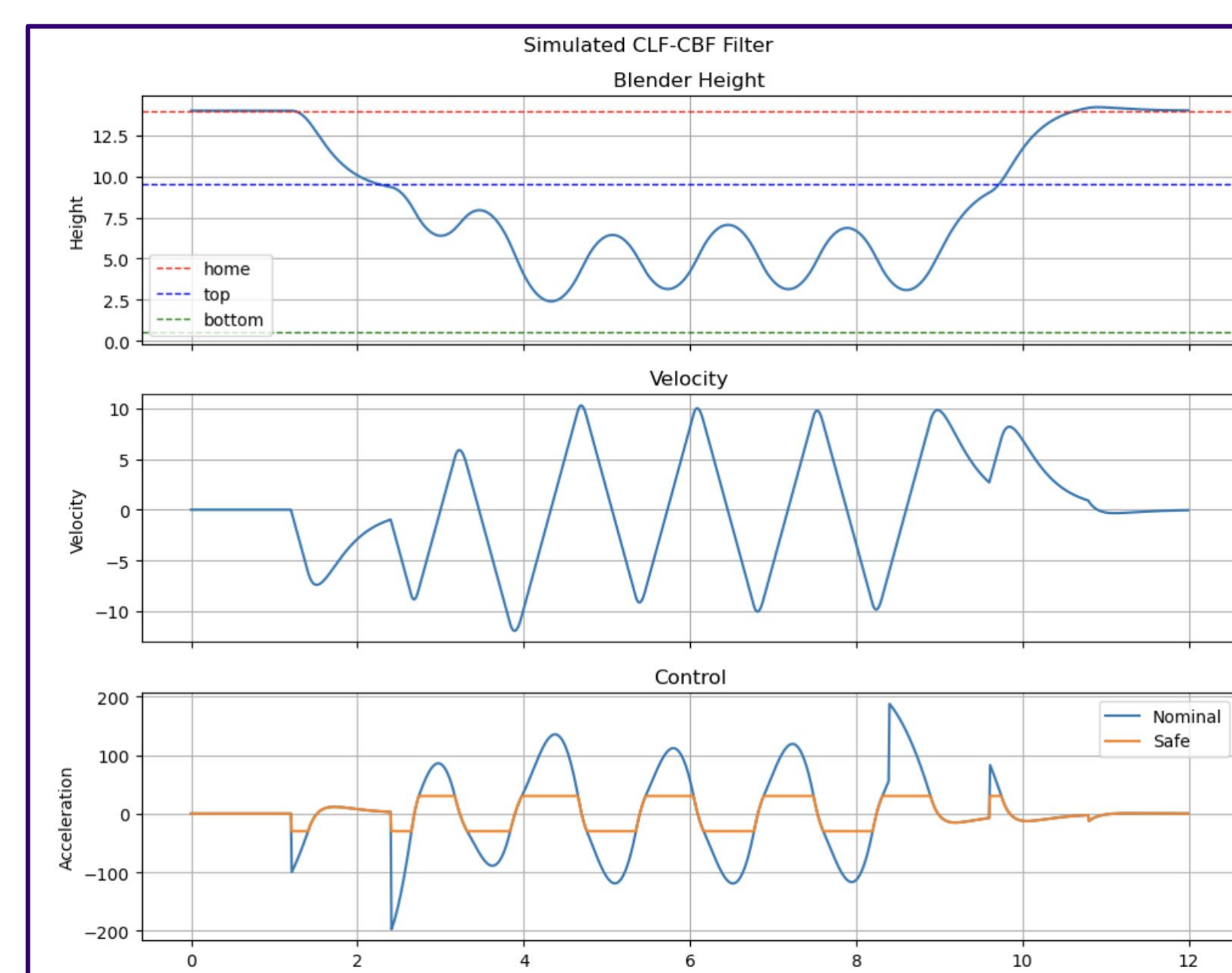


Fig 5: CLF-CBF Filter Simulated Results

Can the CLF-CBF Filter achieve a safe and effective blend profile for a variety of cup sizes?

EXPERIMENTAL RESULTS

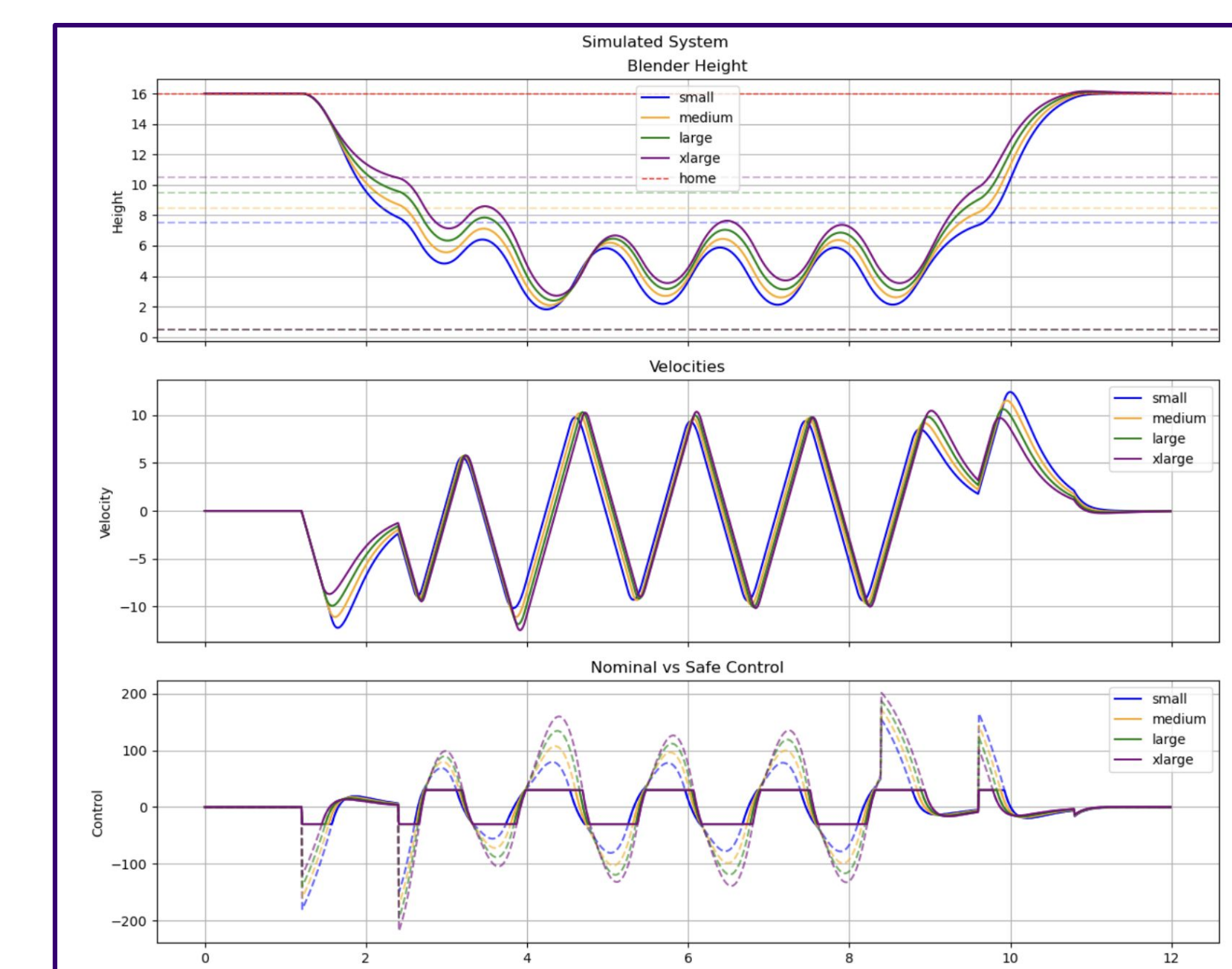


Fig 6: Various Cup Sizes

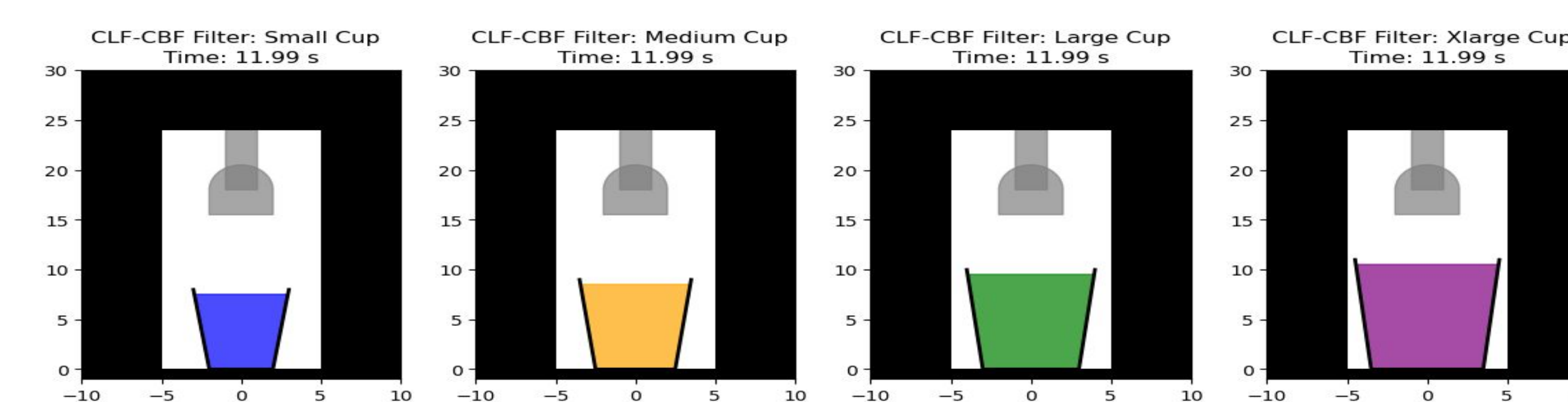


Fig 7: Various Cup Sizes Visualizations

CONCLUSIONS

The CLF-CBF Filter is **applicable** to a variety of cup sizes to **safely guide** the blender tip into the cup, perform the blending action throughout the height of the cup **without hitting the bottom or exiting the cup**, and then **retract safely** to return to the starting position.

FUTURE EXPANSION

Conveyor Belt Extension

- Time horizontal motion of cup to vertical motion of blender decent
- Account for additional cup geometry—top & bottom diameter—to blend deeper into cup while avoiding contact with walls

Additional Opportunities

- Given parameters of drink to be blended, create a learning-based blending profile to perfect blended beverages based on ingredients, etc

References

Ames et al. (2017) – CBF-based Quadratic Programs
Ames et al. (2019) – CBF Theory and Applications
¹Multiplex Beverage, Blend In Cup
Singletary et al. (2022) – Safety-Critical Kinematic Control
Leung – Control Certificate Functions

